PAPER_727: UQFF Quantum Variable Documents: rho_vac UA, rho_vac Ui, v_SCm, rho_vac SCm

Class: `UQFFKnowledgeBaseKB13` **CP4 Entry:** #311 **Keywords:** UQFF, vacuum energy density, rho_vac UA, rho_vac SCm, v_SCm, di-pseudo-monopole, SCm UA ratio **Session:** 176 | **Version:** v5.33 **Source:** grok_share_ba508f76c8e.txt

Abstract

Six quantum variable documents (including duplicates) define the fundamental UQFF vacuum energy density hierarchy: $\rho_{vac}[UA] = 7.09 \times 10^{-36}$ J/m^3 (Universal Aether base), $\rho_{vac}[Ui] = 7.09 \times 10^{-37}$ J/m^3 (Inertia density), $v_{SCm} = 3 \times 10^5$ m/s (SCm flow velocity), $\rho_{vac,A} = 1.683 \times 10^{-10}$ J/m^3 (effective product density), and $\rho_{vac}[SCm] = 7.09 \times 10^{-37}$ J/m^3 (Schwarzschild condensate density). The calibrated ratio $\rho_{SCm}/\rho_{UA} = 0.1$ is fundamental to all UQFF field calculations.

1. Vacuum Density Hierarchy

Density	Value (J/m^3)	Framework layer
$\rho_{vac}[UA]$	7.09×10^{-36}	Universal Aether base
$\rho_{vac}[Ui]$	7.09×10^{-37}	Universal Inertia
$\rho_{vac}[SCm]$	7.09×10^{-37}	Schwarzschild condensate
$\rho_{vac,A}^{eff}$	1.683×10^{-10}	Aether energy product

$$\frac{\rho_{SCm}}{\rho_{UA}} = \frac{7.09 \times 10^{-37}}{7.09 \times 10^{-36}} = 0.1$$

2. SCm Condensate Flow

$$v_{SCm} = 3.0 \times 10^5 \text{ m/s} \approx 10^{-3}c$$

Enters as Doppler-like correction to U_m:

$$U_m^{corrected} = U_m \left(1 + \frac{v_{SCm}}{c}\right)$$

3. Effective Aether Energy Product

$$E_{aether}^{eff} = \rho_{vac,A} \cdot V_{eff} = 1.683 \times 10^{-10} \text{ J}$$

Used in Earth-Moon tidal analogy (KB2/PAPER_717).

References

- UQFF KB grok_share_ba508f76c8e.txt entry #77
- Session 176, v5.33

Whitepaper auto-generated by _gen_whitepapers_716_730.py – Star-Magic Session 176